

## Project Initialization and Planning Phase

|               |                                                     |
|---------------|-----------------------------------------------------|
| Team ID       | MCA                                                 |
| Project Title | Strategic Employee Attrition Intelligence Dashboard |
| Maximum Marks | 3 Marks                                             |

### Project Proposal (Proposed Solution) template

This project proposal outlines a machine learning solution to address an escalating workforce turnover problem. With a clear objective, defined scope, and a concise problem statement, the proposed solution details the methodology, key analytical features, and data-specific resource requirements.

| Project Overview  |                                                                                                                                                                                                                                                                              |
|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Objective         | Build a supervised machine learning classification framework to identify key behavioral predictors of employee attrition and generate real-time turnover risk scores across departments.                                                                                     |
| Scope             | Limited to exploratory data analysis (EDA), statistical feature importance mapping, and predictive model deployment using the historical 5,000-employee HR Information Systems (HRIS) dataset.                                                                               |
| Problem Statement |                                                                                                                                                                                                                                                                              |
| Description       | The organization has suffered unmanaged turnover, losing <b>895 out of 5,000 employees (17.9% attrition rate)</b> . HR currently lacks visibility into whether these exits are driven by compensation gaps, high workload/burnout, poor engagement, or work-mode mismatches. |
| Impact            | Failure to isolate these key attrition drivers prevents HR from prioritizing its retention budget effectively, leading to soaring replacement recruitment costs, loss of institutional knowledge, and compromised team velocity.                                             |
| Proposed Solution |                                                                                                                                                                                                                                                                              |
| Approach          | Train binary classification algorithms (Logistic Regression, Random Forest, XGBoost) to map the target variable (attrition) against multidimensional dataset vectors including compensation, workload, arrangement, and progression scores.                                  |
| Key Features      | <ul style="list-style-type: none"> <li>• <b>Turnover Risk Scorecard:</b> Assigns localized risk probabilities to individual</li> </ul>                                                                                                                                       |

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|  | employee profiles.<br>• <b>Driver Isolation Matrix:</b> Rank-orders top operational predictors (e.g., Overtime Hours vs. Income).<br>• <b>Spend Simulator:</b> Models the theoretical financial ROI of targeted retention interventions. |
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## Resource Requirements

| Resource Type           | Description                                            | Specification/Allocation                               |
|-------------------------|--------------------------------------------------------|--------------------------------------------------------|
| <b>Hardware</b>         |                                                        |                                                        |
| Computing Resources     | Local or Cloud Modern Modern CPU Infrastructure        | Minimum 4-Core современ x86 Modern Architecture        |
| Memory                  | System RAM allocation                                  | Minimum 8 GB RAM                                       |
| Storage                 | Local disk storage for raw tables and model files      | 50 GB Solid-State Drive (SSD) space                    |
| <b>Software</b>         |                                                        |                                                        |
| Frameworks              | Machine learning algorithm and model engines           | scikit-learn, XGBoost                                  |
| Libraries               | Data manipulation, feature scaling, and math utilities | pandas, numpy, matplotlib, seaborn                     |
| Development Environment | IDE workspace and version control system               | Jupyter Notebook / VS Code, Git                        |
| <b>Data</b>             |                                                        |                                                        |
| Data                    | Internal corporate HR Information Systems dataset      | 5,000 unique records, 29 distinct features, CSV format |